

JKBOSE 2026

CLASS 11TH

#XUV ❤️

Bhaiichara ON
TOP.

CHEMICAL BONDING AND MOLECULAR STRUCTURE

LECTURE-3

CHEMISTRY

VISHAL SIR



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Topics To Covered

- 1 Bond Parameters ✓
- 2 Calculation of formal charge. ✓
- 3 Limitations of Octet Rule ✓



REVISION.

① Factors affecting formation of ionic bond.

- IE $\downarrow$
- Δ_{eg} highly -ve.
- Lattice energy $\uparrow$

Covalent bond $\Rightarrow$



Lewis dot structure $\Rightarrow$

i) TVE =

ii) Central atom $\Rightarrow$

iii) Total no. of bonds.

$$\frac{\text{Req. } e^- - \text{TVE}}{2} =$$

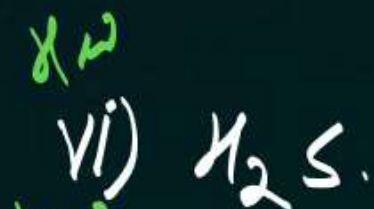
Writing Lewis Structures or Electron Dot Structures



EXAMPLE:



s2.



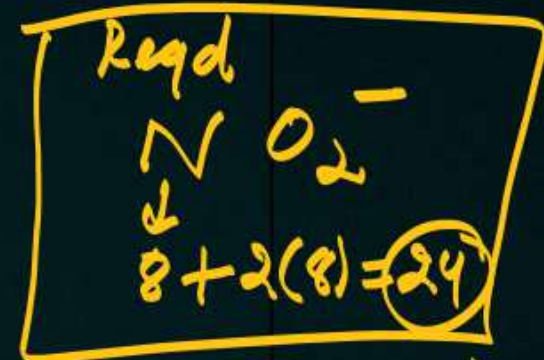
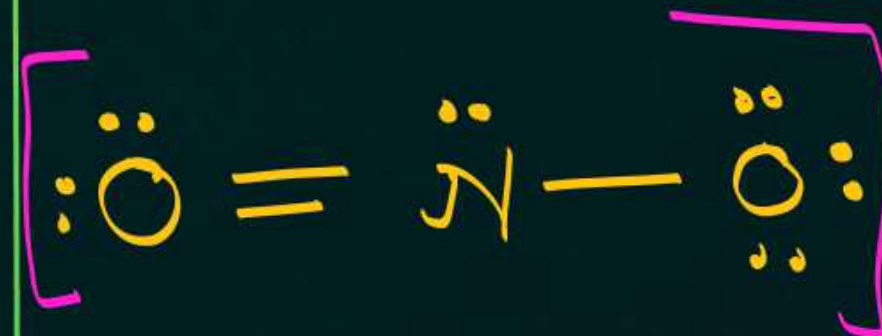
$$\text{TVE} = 5 + 2(6) + 1 = 18\bar{e}$$

Central atom = Nitrogen.

Total no. of bonds $\rightarrow$

$$\frac{\text{Reqd. valence } \bar{e} - \text{TVE}}{2} = \frac{24 - 18}{2}$$

$$= \frac{6}{2} = 3$$



IV) O_3 .

$$TV\epsilon = 3(6) = 18e$$

Central atom = Oxygen.

$$\text{Total no. of bond} = \frac{\text{Reqd. Valen } e - TV\epsilon}{2}$$

$$= \frac{3(8) - 18}{2} = \frac{24 - 18}{2} = \frac{6}{2} = 3$$

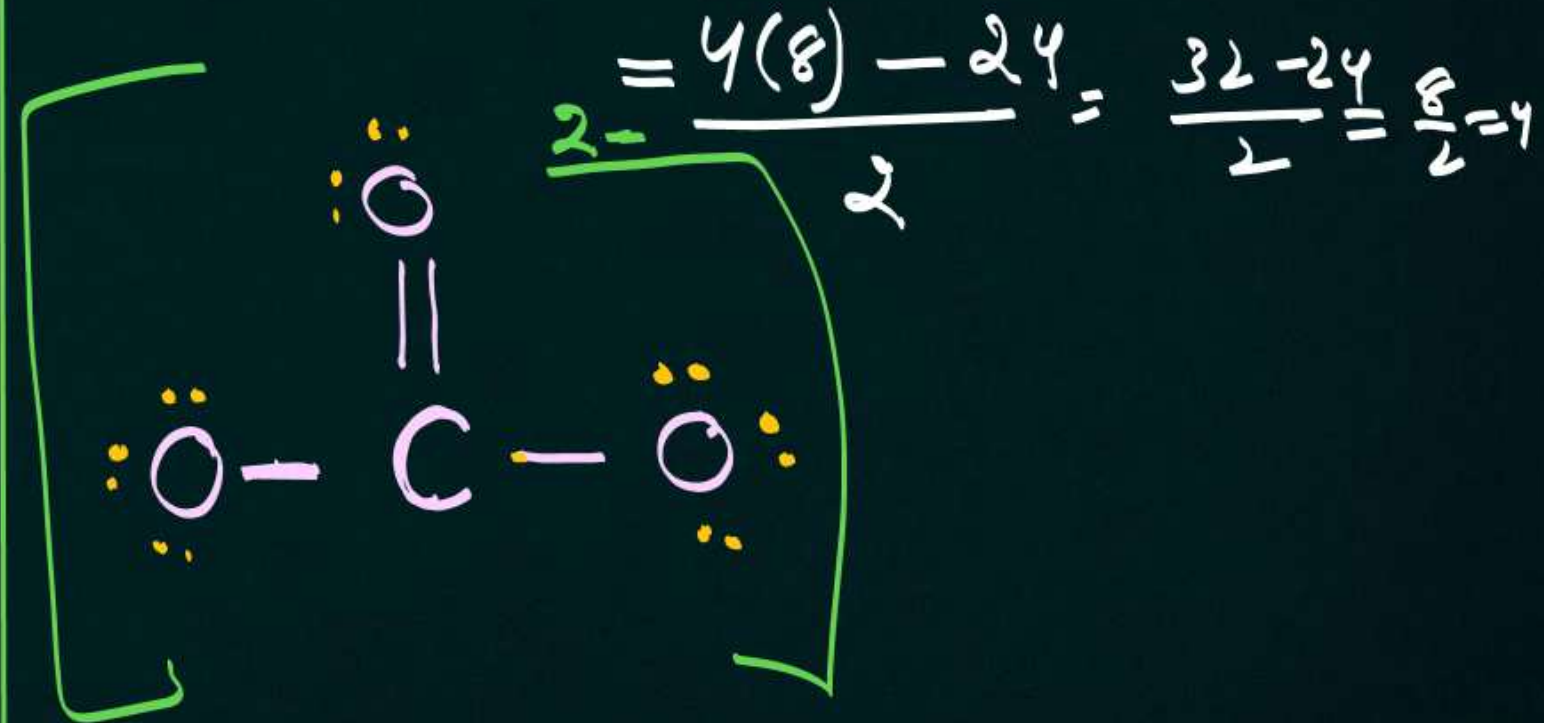


viii) CO_3^{2-} (carbonate ion).

$$\rightarrow TV\epsilon = 4 + 3(6) + 2 = \underline{24e}$$

$\rightarrow$ Central atom: $\rightarrow$ Carbon.

$$\rightarrow \text{Total no. of bonds} \rightarrow \frac{\text{Reqd. Valen } e - TV\epsilon}{2}$$





$$TVE = 1 + 4 + 6 + 6 + 1 = 18e^-$$

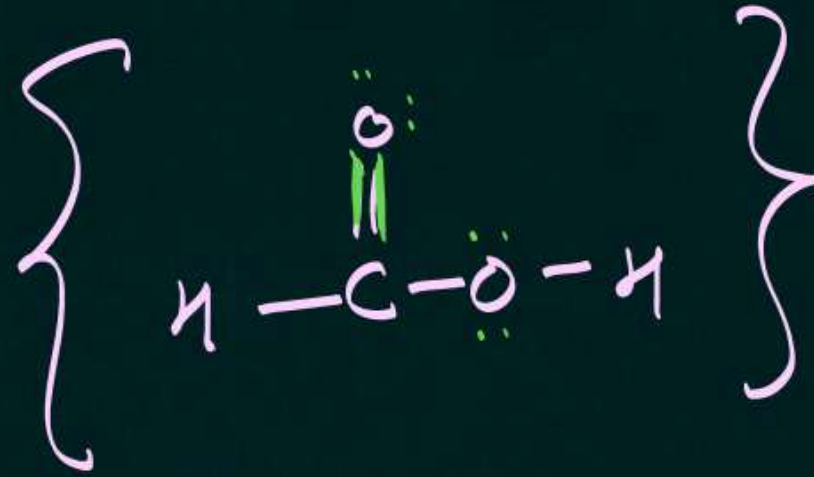
Central atom = Carbon.

Total no. of bonds $\Rightarrow$
$$\frac{\text{Required V.E} - TVE}{2}$$

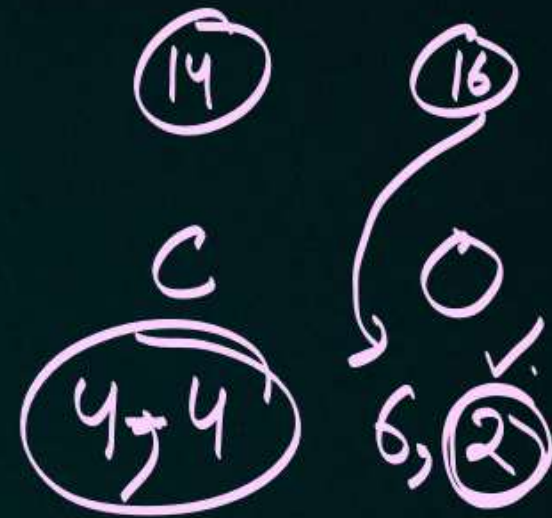
$$= \frac{(2 + 8 + 8 + 8 + 2) - 18e^-}{2}$$

$$= \frac{28 - 18}{2} = \frac{10}{2} = 5$$

NEAT ✓



Valency.



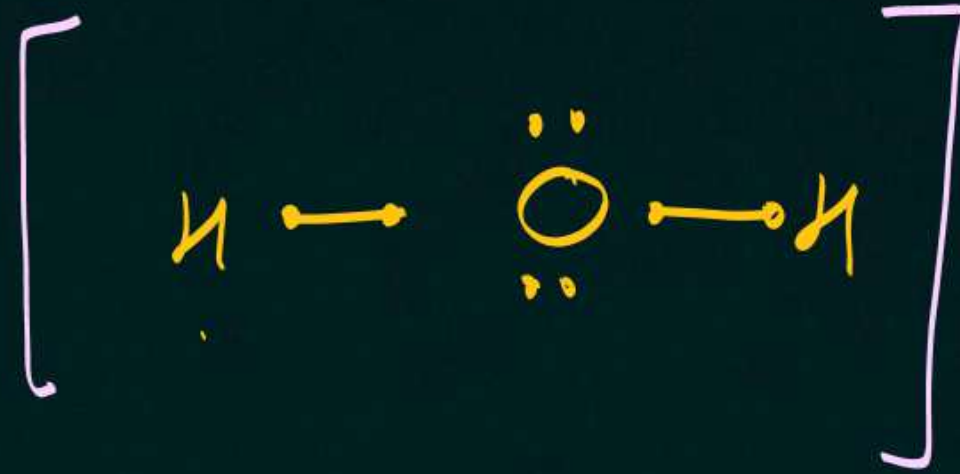
iv). H_2O .

$$TVE = 2 + 6 = 8e^-$$

Central atom $\rightarrow$ Oxygen.

$$\text{Total no. of bonds} \rightarrow \frac{\text{Reqd. V.E} - TVE}{2}$$

$$= \frac{(4 + 6) - 8e^-}{2} = \frac{12 - 8}{2} = \frac{4}{2} = 2$$



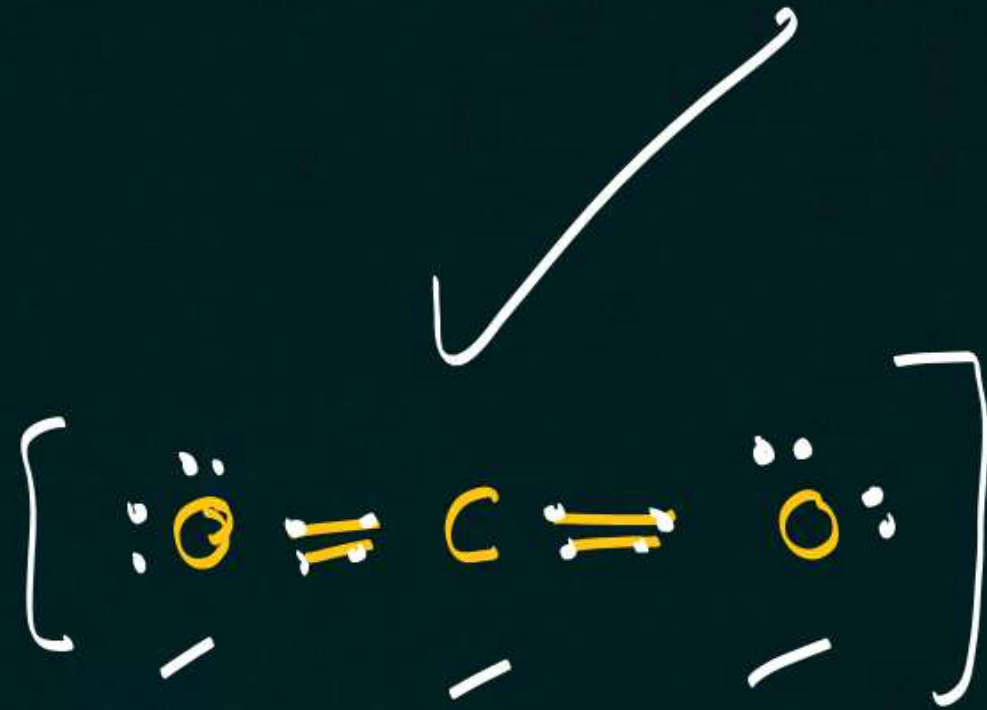
i) CO_2

$$\text{TV} \varepsilon = 4 + 2(6) = \underline{16 \varepsilon}$$

Central atom = Carbon.

$$\text{Total no. of bonds} = \frac{\text{Reqd. Valence } \varepsilon - \text{TV} \varepsilon}{2}$$

$$= \frac{(8 + 16) - 16}{2} = \frac{24 - 16}{2} = \frac{8}{2} = \textcircled{4}$$

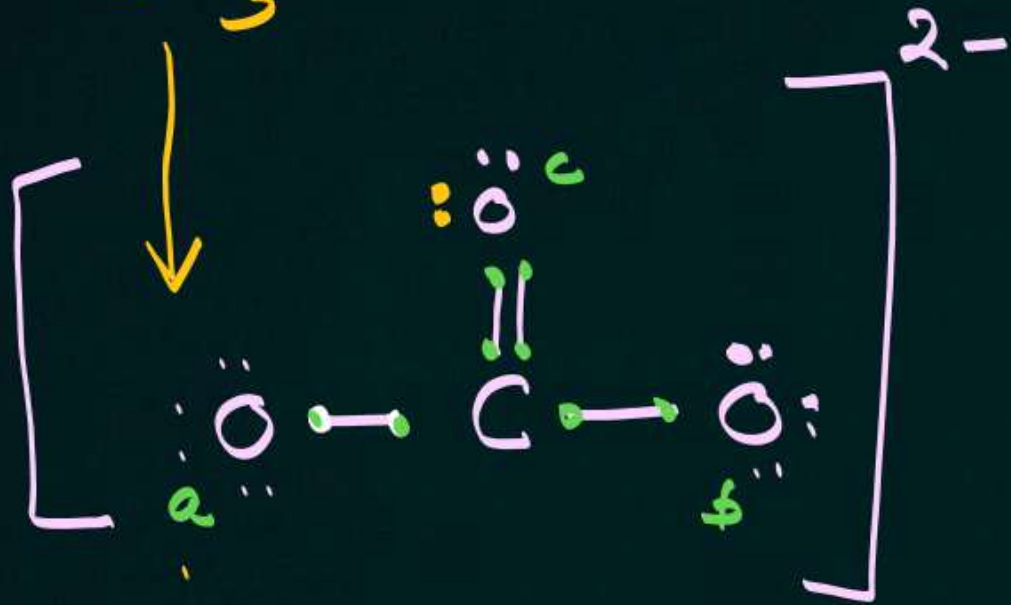


(duplet)
H $\rightarrow$ 2 ε for stability

$\downarrow$ $\textcircled{8 \varepsilon}$ Reqd. $\rightarrow$ octet Rule.

Calculation of formal charge $\Rightarrow$

e.g. CO_3^{2-}



-ve charge $\rightarrow$ Side atom
+ve charge in Central atom.

$$\begin{aligned} \text{iv) F.C on Carbon} &= V - \frac{S}{2} - L \\ &= 4 - \frac{8}{2} - 0 \\ &= 4 - 4 = 0. \end{aligned}$$

Formal charge $\Rightarrow$

$$V - \frac{S}{2} - L$$

V = Total valence e^-

S = Total e^- in shared pairs (bond).

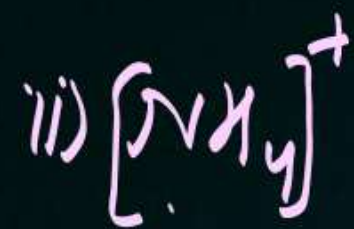
L = Total e^- in lone pair



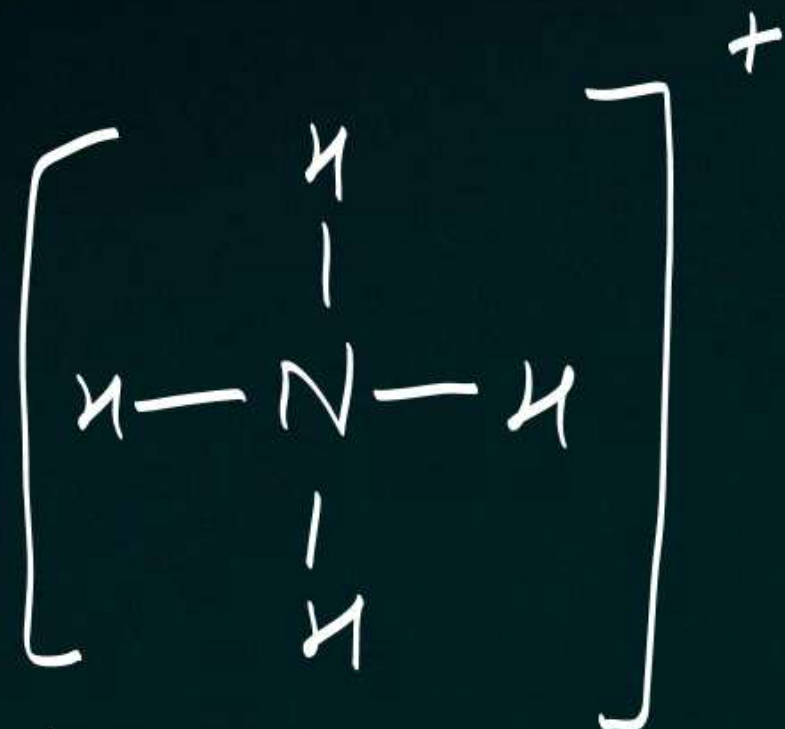
$$\begin{aligned} \text{i) F.C on } O_a &= V - \frac{S}{2} - L = 6 - \frac{2}{2} - 6 \\ &= \cancel{6} - 1 - \cancel{6} = \textcircled{-1} \end{aligned}$$

ii) Same for $O_b = \textcircled{-1}$

$$\begin{aligned} \text{iii) F.C on } O_c &= V - \frac{S}{2} - L = 6 - \frac{4}{2} - 4 = 6 - 2 - 4 \\ &= 6 - 6 = \textcircled{0} \end{aligned}$$



Lewis-dot structure.



Calculate the formal charge on Nitrogen $\Rightarrow$

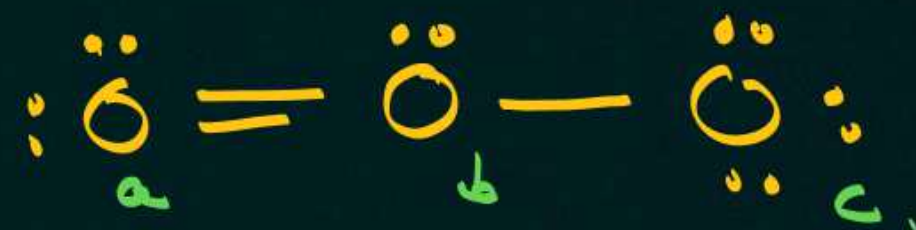
$$\text{F.C} = V - \frac{S}{2} - L = \left[5 - \frac{8}{2} - 0 \right] = 5 - 4 = (+1)$$

Now

Q. Calculate the formal charge on each oxygen atom of O_3 .

Sol $\Rightarrow$

Lewis dot structure



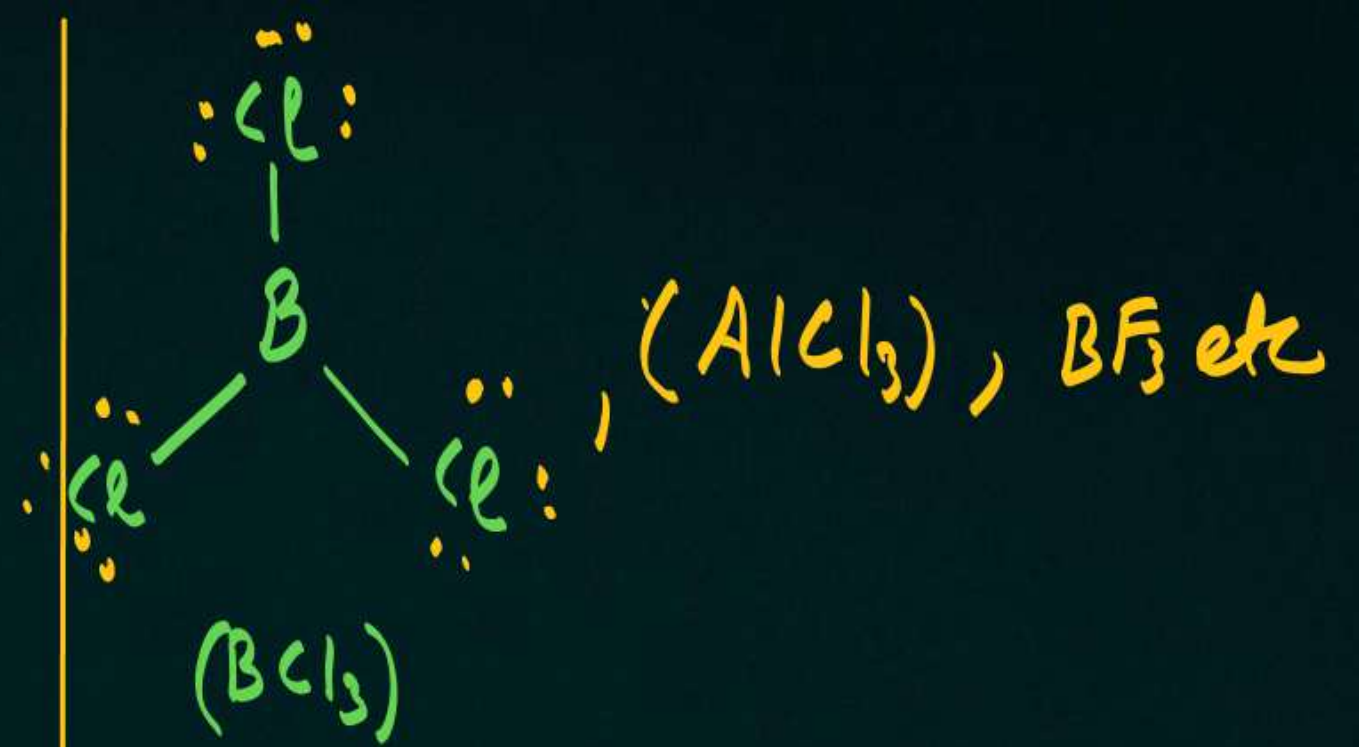
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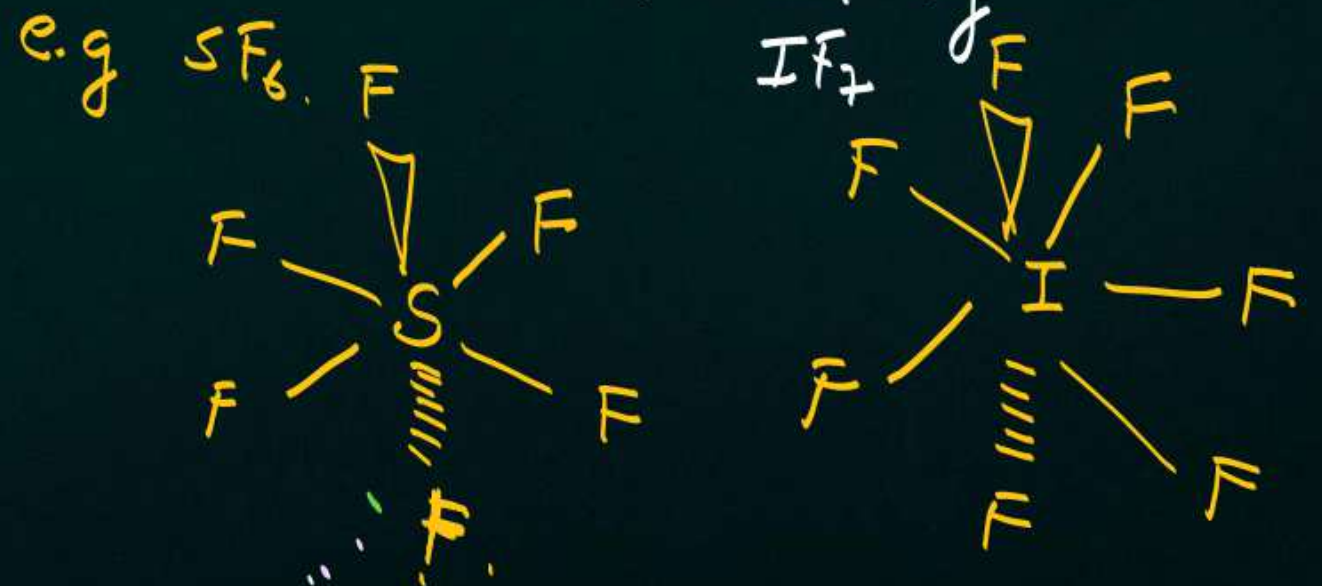
LIMITATIONS OF OCTET RULE.

i) Compounds of hydrogen $\rightarrow$ Hydrogen has one e^- in their valence shell ($1s^1$) & require only one e^- to achieve nearest noble gas configuration hence hydrogen follow duplet not octet.

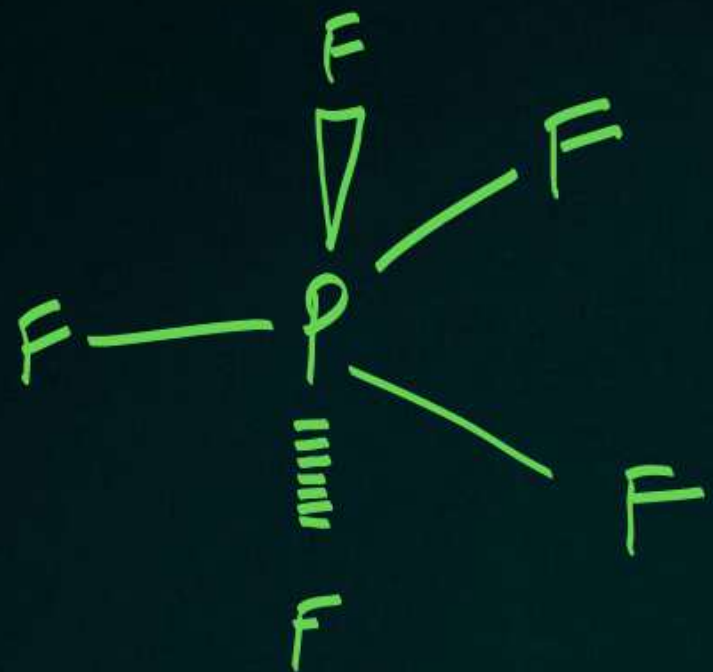
ii) Hypovalent Species $\rightarrow$ Compounds in which central atom has less than eight valence electron but still stable. e.g



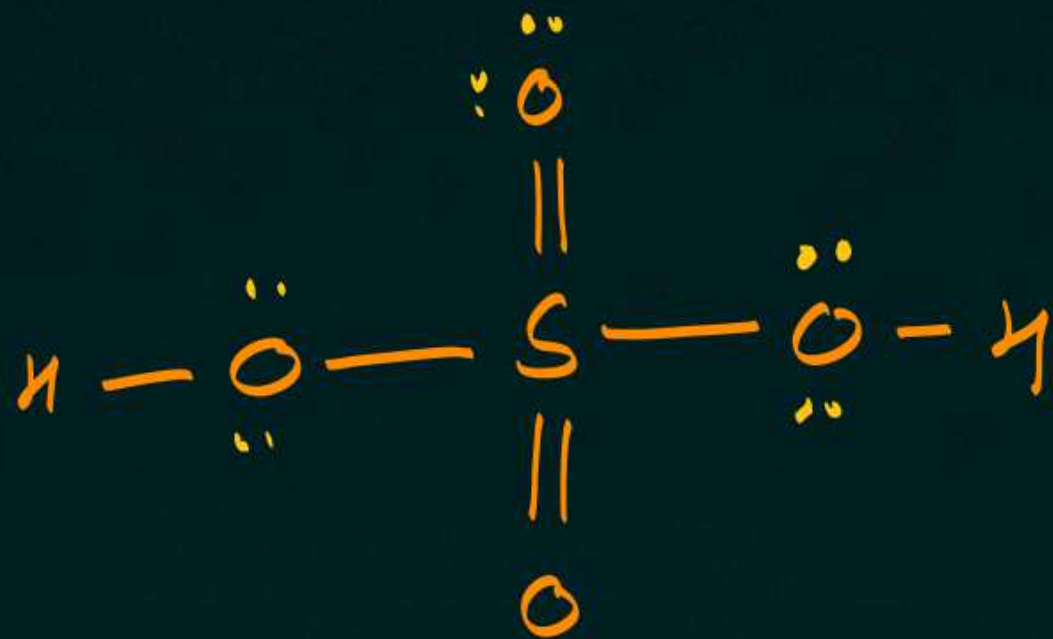
iii) Hypervalent Species $\rightarrow$ Compounds in which central atom has more than eight e^- in their valence shell. e.g



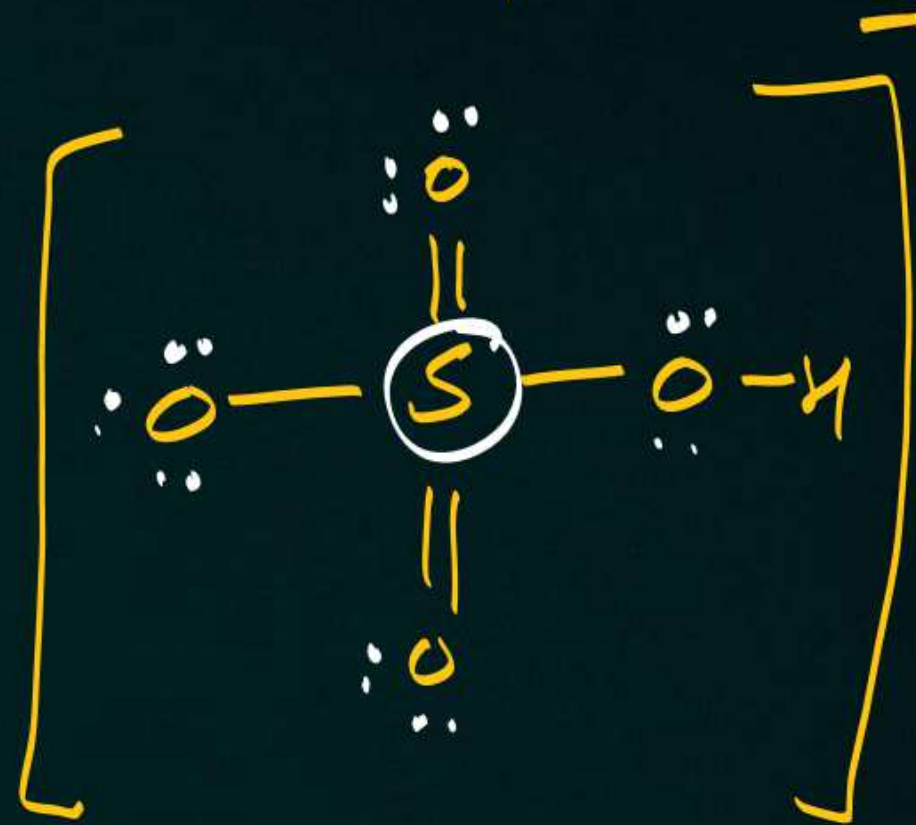
PCl_5 or PF_5



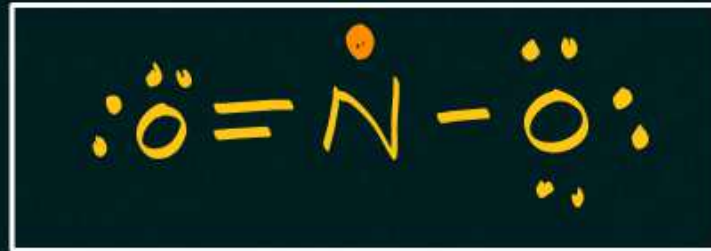
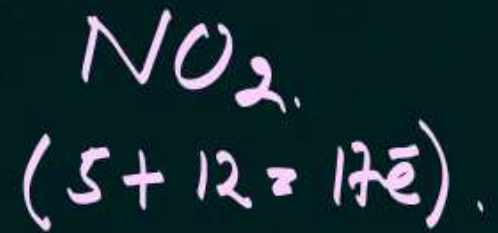
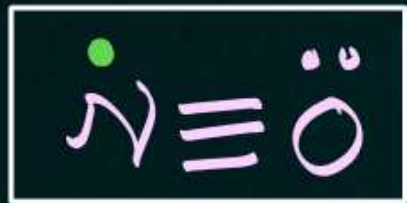
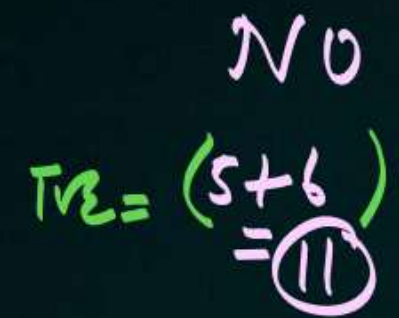
, * H_2SO_4 .



HSO_4^-



IV) Odd $\bar{e}$ species $\rightarrow$ Compounds have odd $\bar{e}$ do not follow octet Rule. e.g.



V) Compounds of Nobel gases $\rightarrow$
 Few noble gases also form compounds.
 e.g. KrF₂, XeF₄, XeO₄ etc.

Valence $e^- \rightarrow$ ^{Group no.}

Reqd. v-e = $8e^-$ or $(2e^-)$




QUESTION



What is the total number of valence electrons in a molecule of NF_3 ?

1 8

2 16

3 26

4 32

hallo sawal

TVΣ in NF = ✓.



$$5 + 3(\underline{7}) = 5 + 21 = \textcircled{26}$$

QUESTION

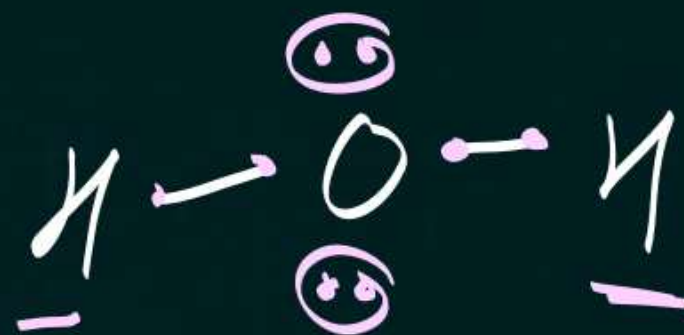


How many lone pairs of electrons are present on the oxygen atom in a molecule of water (H_2O).

Lalun sawal.

lp = ?

Total $\bar{e}$ in lp X



1

1

2

2

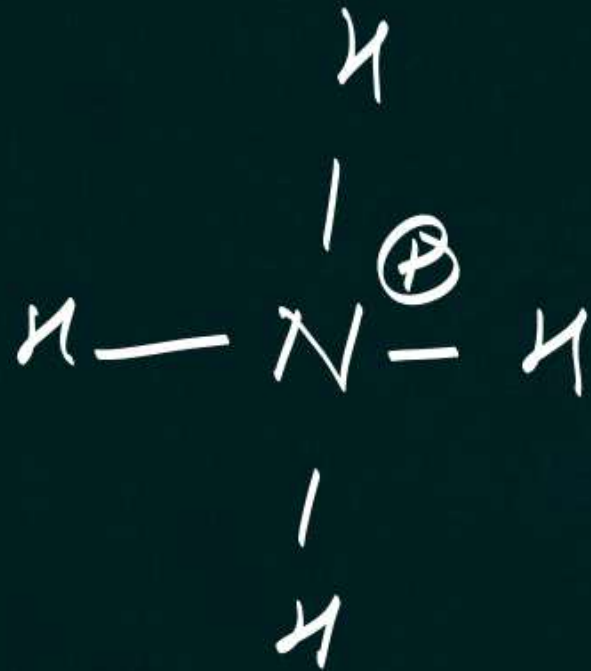
3

3

4

4

Q. Which of the following follow Octet Rule $\rightarrow$



Thank
You